

Investigation Log: Periodicity of the Octal Game 0.45

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Abstract

Following an audit that found four consecutive candidate results in low-dimensional topology already claimed by an active research community (see `manuscript/log.tex`, Log Entries 17, 19, 23), this log documents a deliberate pivot to a different mathematical neighborhood: combinatorial game theory, specifically the still-open general conjecture that every finite octal game has an eventually periodic Sprague–Grundy sequence. A specific, small, unresolved instance (the octal game 0.45) is identified, its periodicity conjecture (preperiod 498, period 20, per A. Flammenkamp’s published database) is independently reproduced from scratch, and a real structural mechanism explaining the period is found and extensively verified: the sequence is confined to a 5-element value set beyond $n = 198$, and the recursion determining each term has an empirically bounded effective depth far smaller than its rules a priori permit. This is reported at the resolution it has actually reached: strong, original, extensively-verified structural progress, not a completed formal proof.

Entry 1: Why This Problem

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After the knot-theory audit’s fourth consecutive miss (`manuscript/log.tex` Log Entry 23), a genuinely different area was sought, following the same standard applied throughout: find a specific, well-defined, checkably-open target before investing effort, rather than assume openness.

The area. Octal games (Guy–Smith notation $0.d_1d_2d_3\dots$) are simple take-and-break Nim variants; whether every finite one has an eventually periodic nim-sequence is a major open conjecture in the field. Of 167 inequivalent octal games with at most 3 code digits, only 20 have been rigorously proven periodic despite millions of computed nim-values by A. Flammenkamp, who maintains the field’s reference database.

The candidate. 0.45: remove 1 token from a heap only by splitting the remainder into two nonempty heaps (needs heap ≥ 3); remove 2 tokens either to end the heap (heap = 2) or by splitting the remainder (needs heap ≥ 4). Flammenkamp’s database reports a conjectured preperiod of 498 and period of 20 – by a wide margin the smallest such pair among the unresolved small-code games checked (.356: 7315/142; .156: 3479/349; .644: 3256/442 – all independently verified, see below). 0.45 is not among the four games (.127, .16, .376, .56) already proved periodic by Gangolli–Plambeck; no published proof for it was found.

Entry 2: A Validated Engine

Date: September 23, 2026 **Command:** `python src/octal_games/grundy.py src/octal_games/grundy.py` computes Sprague–Grundy sequences for arbitrary octal codes from scratch (a direct mex computation, no special-casing). It is validated two ways:

- **Kayles (0.77):** the engine’s own search finds preperiod 71, period 12 – an **exact match** to R. K. Guy’s 1949 hand-proved result, independently cited in secondary literature.
- **Four cross-checks against Flammenkamp’s database:** 0.45 (498, 20), 0.356 (7315, 142), 0.156 (3479, 349), 0.644 (3256, 442) – all four reproduced exactly by this engine before being trusted for anything further, after an earlier AI-search summary of the same database gave a contradictory status for a related game (.16) that turned out to be wrong on inspection of the primary source.

Full output: `results/octal_games/grundy_selftest.txt`.

Entry 3: Three Findings on 0.45

Date: September 23, 2026 **Command:** `python verification/investigate_octal_45.py`
All computed up to $N = 60,000$ ($300\times$ the last exception found; roughly 3000 periods past the conjectured onset).

1. Boundedness. $G(n) \in \{1, 2, 4, 7, 8\}$ for all $199 \leq n \leq 60,000$, with zero exceptions. Below $n = 199$ there are exactly 11 exceptions ($n = 0, 1, 6, 11, 19, 31, 78, 88, 98, 106, 198$), and Flammenkamp’s own published bitmask for this game is all-1s, so his “sparse/rare value” criterion (`popcount(G(n) AND m) AND 1 == 0`) reduces exactly to “`popcount(G(n))` is even” – which is exactly this set of 11 exceptions and nothing else, matching his database’s reported count and largest-rare-index exactly.

2. Stabilization. The mex-determining reachable set at position n draws, in principle, on splits $a + b = n - 1$ (or $n - 2$) for every $1 \leq a \leq \lfloor (n - 1)/2 \rfloor$ – a range that grows without bound as n grows. Empirically, restricted to the values $\{0, \dots, 8\}$ that can actually affect the mex (given finding 1), the reachable set is fully determined by only the first A split terms, for a bound A found to be at most 98 across 416 sampled $n \in [500, 60,000]$ (worst case $A = 78$ at $n = 526$ in a focused re-check). Terms beyond A – checked out to $a \sim 30,000$ for some n – contribute nothing new.

3. Closure. Combining 1 and 2: for n large enough that its bounded window of relevant recent terms lies entirely past $n = 199$ (i.e. in the fully-bounded regime), the reachable set restricted to $\{0, \dots, 8\}$ – and hence $G(n)$ itself – depends only on $n \bmod 20$. Checked directly, with zero exceptions, across thousands of $(n, n+20)$ pairs spanning the full computed range.

Entry 4: Honest Status

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This is **not** a completed formal proof that 0.45’s nim-sequence is periodic for all n . What it is:

- An original structural account – not previously found in any source consulted – of *why* the period-20 pattern holds: a bounded value set plus a bounded effective recursion depth, rather than an unexplained empirical repeat.
- Verification far beyond the standard of evidence usually meant by “observed periodic” in this literature: 300× the last exception, with the stabilization bound checked (not merely assumed) across hundreds of large, widely-spaced sample points.
- A concrete, stated route to a complete proof: show the stabilization bound (Finding 2) holds for *all* n past some explicit threshold, directly from the recursion’s definition – which would, combined with Finding 1, close the argument by the standard periodicity-certification method used throughout this literature (verify one clean window, then propagate by the bounded-depth structure).

This is reported at exactly this resolution, per the same policy stated in `manuscript/log.tex`
 Log Entry 3: distinguish clearly between what has been shown and what is strongly evidenced but not yet closed, and do not round up.